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Class 9
Chemistry
Chapter 4
NOTES

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"Periodic table & Periodicity of Properties"

Question & Answer:-

Question No 1:- Define Periodic table?

Answer:- Periodic table:- A table showing systematic arrangement of elements is called periodic table.

Question No 2:- What is Periodic Law?

Answer:- Periodic ^{Law} state that if the elements are arranged in the order of their increasing atomic number, their properties are repeated in a periodic manner.

Question No 3:- How many period are there in Periodic table?

Answer:- Period:- The horizontal rows of the periodic table are called period.

The number of elements per period range from 2 in period 1 to 32 in period 6. First three periods are called short periods & the remaining periods are called long periods. The properties of elements within a period change gradually as you move from left to right in it. But when you move from one period to the next, the pattern of properties within a period repeats. This is an occurrence to the periodic law.

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Question no 4: How many groups in the Period table?

Answer: Groups: Each vertical column of elements in the Periodic table is called group or family.

The first 2 groups are IA & IIA, while the last 6 groups are IIIA to VIIIA & the middle groups are in group B. The elements in the same group have same number of valence electron.

For example, group 1 & 2 elements have 1 & 2 valence electrons respectively. In group 13 element have 3, group 14 have 4, group 15 have 5 valence electrons & so on.

Group A elements are called normal or representative elements. They are also called main group elements. Group B elements are called transition elements.

Add in it group IA elements are called "alkali metals"

group IIA elements are called "alkaline Earth"

group VIIA elements are called "Halogens"

group VIIIA elements are called "noble gas"

Question no 5: Define electronic Configuration.

Answer: Electronic Configuration:

According to Aufbau's principle the order in which the orbitals fill up is as follows:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p & so on.

Each orbital has a fixed capacity for the maximum no. of electrons accommodated. s-orbitals have the capacity of 2 electrons, while p orbitals have the capacity for 6 electrons, d orbitals have the capacity for 10 electrons & f orbitals have the capacity for 14 electrons.

	IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
Period 1:	1	R						
Period 2:	T	S	Q	P				W
			Z					
Period 3:				X				Y

Question no 6: What are the s & p blocks in period table.

Answer: Group 1 & 2 elements contain their valence electrons in the s-sub-shell. Therefore these elements are called s-block elements. Elements in group 13 to 18 (except He) are known as p-block elements because their valence electrons are located in the p-sub-shell. Lanthanides & actinides are known as f-block elements since their valence electrons lie in f-sub-shell. Figure 4.21 shows the blocks of the Period table.

$\text{Li} = 1s^2 2s^1$, as valence electron is in s-sub-shell, Li belongs to s-block.

$\text{C} = 1s^2 2s^2 2p^2$, as valence electron is in sub-shell p, C belongs to p-block.

Question no 7: Describe group number change on an "ion"

Answer: The group number of an s-block element, the period table corresponds to its no. of valence electrons. No. of valence electrons equal to group no. minus 10.

Whereas in the case of p-block elements, the number of valence electrons is equal to group no. minus 10. Some elements tend to lose electrons. Why? Elements tend to achieve a stable electron configuration such as the noble gas. Remember that the 2 or 8 electron configuration is the most stable configuration. Elements with 1-3 electrons in their valence shell tend to lose those electrons & form +1, +2, +3 ions respectively & form negatively charged

most stable configuration. Elements with 1-3 electrons in their valence shell tend to lose those electrons & form +1, +2, +3 ions respectively & form negatively charged

ion with -3, -2, -1 charges, respectively. Elements with 4 valence electrons can lose 4 electrons to form +4 ions. They are also known to gain 4 e⁻ & form -4 ions.

*** Group I (Alkali metals):** Group elements such as Lithium (Li), Sodium (Na) & Potassium (K) have one valence electron & belong to s-block. s-block elements lose electrons equal to their group no. They tend to lose this electron to form a +1 ion, also known as a monovalent cation.

Eg: Lithium (Li) loses one electron to form Li^+ Sodium (Na)

*** Group II (Alkaline earth metals):**

Group II elements such as Be, Mg, Ca have two valence electrons. They tend to lose these two electrons to form X^{+2} ion

Eg:- Be^{+2} , Mg^{+2} , Ca^{+2}

*** Group 17 elements (Halogens):**

Group 17 elements such as F, Cl, Br, I have seven valence electrons. They tend to gain one electron to form X^{-1} ion & achieve noble gas configuration.

Eg:- F^{-1} , Cl^{-1}

Group 16 (Chalcogens):

Group 16 elements such as O, S have six valence electrons. They tend to gain 2 electrons to achieve noble gas configuration & form X^{-2} ion

Eg:- O^{-2} , S^{-2}

Group 18 (Noble gases):

These elements such as He, Ne & Ar have valence electron shell. They are inert gases due to stable electronic configuration.

Question No 8:- What is Periodicity of Properties?

Answer:- Elements with a similar Valence shell electronic Configuration are placed in the same group, chemical properties depend on the electronic Configuration of the Valence shell. Because all the elements in a given group have similar valence shell electronic Configuration. Physical Properties depend on the Size of atoms. Because the sizes of atoms change gradually from top to bottom in a group. Therefore, the elements show a gradation of Physical properties within the same group. The number of electrons in the valence shell increases gradually from left to right. Their chemical & physical properties also differ in the same way.

Question No 9:- Define shielding effect, write about trends in the Period table.

Answer:-

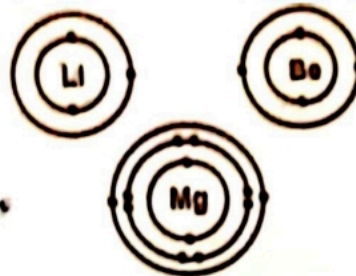
The reduction in Force of attraction b/w nucleus & the valence electron by the electrons present in the inner Sub-shell is called shielding effect.

Electrons present in the inner shell cut off attractive force b/w the nucleus & the valence electrons.

i) Trends in groups:-

As you move from top to bottom in a group the no of electrons shell increases. So the number of electrons in the inner shell also increase. As a result shielding effect increase.

Electronic structure
of Li, Be and Mg



Trends in Periods:-

As you move from left to right in a period table the no of electrons in the inner shells remains constant therefore, shielding effect remains constant.

Question No 10 :- Define atomic size, what are its trends in Period Time?

Answer:-

Atomic size is the average distance b/w the atomic nucleus & the electronic outer shell.

Trends along the Periods:-

The atomic radius decreases in each period. This is because, Another, electron is added to the same valence shell. At the same time, the positive charge of the core also increases by one. The attraction of the nucleus to the electron in the valence shell increases. Therefore, the size of the shell & the radius of the atom decreases.

Trends along the groups:-

Atomic radius increases in each main group as you move down the element group. As you move down the group to the next lower element, the atom has an additional shell of electrons. This increases the radius of the atom.

Group 16 elements such as O, S have 6 valence electrons.

They tend to gain 2 electrons to achieve noble gas configuration & form x^{-2} ion.

Eg:- O^{-2}, S^{-2}

Group 18 (Noble gases):

These elements such as He, Ne & Ar have valence electron shell

They are inert gases due to stable electronic configuration.

— 08 (Exercise 4.2) 80 —

Obtaining the position of elements in the periodic table from the electronic configuration.

Find out the position of the following elements in the P.T.

i) Nitrogen (Atomic no 7)

Sol Electronic Configuration of N = $1s^2 2s^2 2p^3$

Valence Shell has Configuration: $2s^2 2p^3$

Period number = 2

No of Valence electrons = $2+3=5$

N belongs to p-block

So Group Number = $5+10=15$

Nitrogen is present in the 2nd period of Group 15.

ii) Oxygen (Atomic no 8)

Sol Electronic Configuration of O = $1s^2 2s^2 2p^4$

Valence Shell has Configuration: $2s^2 2p^4$

Period number = 2

No of Valence electrons = $2+4=6$

O belongs to p-block

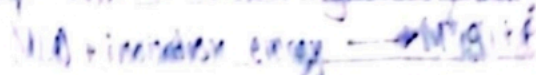
So Group Number = $6+10=16$

Oxygen is present in the 2nd period of Group 16.

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Question:- What is ionization energy? Explain its trends in Periodic table?

Answer:- Definition: "Ionization energy is defined as the minimum amount of energy required to remove the outermost electron from an isolated gaseous atom."



The unit of ionization energy is 'kJ mol⁻¹' or 'eV atom⁻¹'.

Trends along the groups:-

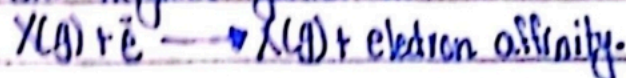
The value of the ionization energy decreases from top to bottom in the group. This is because the shielding effect of the inner electrons increases down the group.

Trends along the periods:-

As you move from left to right in the periods, the shielding effect remains unchanged. But little by little the nuclear charge increases. So, stronger attraction b/w the nucleus & the outer electron increases. As a result, the ionization energy increases from left to right in the period.

Question No. 10:- What is Electron Affinity? Explain its trends in Periodic table?

Answer:- Definition: "Electron affinity is defined as the amount of energy released when an electron adds up in the valence shell of an isolated atom to form a uni negative gaseous ion."



Trends along the Period:-

As you move from left to right through a period, electron affinity generally increases. This is due to an increase in nuclear charge & a decrease in atomic radius.

Trends along the Group:-

Electron affinity decreases from top to bottom in a group. This is due to an increase in the shielding effect. Due to the increased shielding effect & increase in atomic radius, the added electron binds less tightly to the nucleus. As a result, less energy is released.

electron affinities of main group elements									
H -73									He 0
Li -60	Be 0		B -27	C -122	N +7	O -141	68/144		
Na -53	Mg 0		Al -44	Si -134	P -71.7	S -200	Cl -349	Ar 0	
K -48	Ca 0		Ga -29	Ge -120	As -77	Se -195	Br -325	Kr 0	
Rb -47	Sr 0		In -29	Sn -121	Sb -101	Te -190	I -295	Xe 0	
Cs -45	Ba 0		Tl -30	Pb -110	Bi -110	Po -180	At -270	Rn 0	

Ionization energies of the main group elements

	IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
1	H 1312							He 2372
2	Li 520	Be 899	B 801	C 1086	N 1402	O 1314	F 1681	Ne 2081
3	Na 496	Mg 738	Al 578	Si 786	P 1012	S 1000	Cl 1251	Ar 1521
4	K 419	Ca 590	Ga 579	Ge 762	As 947	Se 941	Br 1140	Kr 1351
5	Rb 403	Sr 549	In 558	Sn 709	Sb 834	Te 869	I 1008	Xe 1170
6	Cs 376	Ba 503	Tl 589	Pb 716	Bi 703	Po 812	At 826	Rn 1037

Concept Assessment Exercise no: 4.5

Topic: Shielding effect.

Question:- Choose the element whose atoms you expect to have smaller shielding effect.

i) F or Cl:-

Sol:-

As, we look in the Periodic table F & Cl is present in Group 17

F has less Shielding effect.

Reason:-

As we move down the group Shielding effect will be increases.

ii) Li or Na:-

Sol:-

Li will have less shielding effect.

iii) B or Al:-

Sol:-

B will have Less Shielding effect.

Concept Assessment Exercise No:- 4.6
Topic:- Atomic Size.

Question:- Choose the atom which have less smaller atomic size.

i) O or S

Sol:-

O has less atomic size.

Reason:-

As we move down the group the atomic size will be increase.

ii) O or F

Sol:-

F will have less atomic size.

Concept Assessment 4.7

Topic:- Ionization energy

Question:- which atom have smaller ionization energy.

a) B or N

Sol:-

B will have less ionization energy.

b) Be or Mg

Sol:-

Mg will have less ionization energy.

c) C or Si

Sol:-

Si will have less ionization energy.

— os (Question no and Answer) —

Question 17. What is Electronegativity. Explain

Mg will have less ionization energy.

c) C or Si

Sol: c

Si will have less ionization energy.

— 08 (Question no and Answer) —

Question no 13: What is Electronegativity. Explain its trends in the Period table?

Answer:-

Electronegativity:-

Electronegativity is the ability of an atom to attract electrons towards itself in a chemical bond.

Trends down the group:-

As we move down the group the electronegativity decreases.

Trends along the Period:-

As we move left to right the electronegativity of the element will increase.

H 2.1							He
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne 2.1
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar 3.0
K 0.5	Ca 1.0	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr 2.1
Rb 0.8	Sr 1.0	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe 2.6
Cs 0.7	Ba 0.9	Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.2	Rn
Fr 0.7	Ra 0.9						

n.7 n.9

Question No 14:- What is metallic character. Explain its trends in Period table?

Answer:- Definition:- Metallic nature refers to a property of elements in the periodic table that determines how easily they can lose electrons & form positive (cation) Elements with metallic character have a strong tendency to lose electrons & easily form cations.

Trends down the group:-

Metallic character increases as you move down a group in the Period table. This is primary due to addition of new electronic shell. The outermost electrons are further from the nucleus & experience weaker attractive forces.

Trends along the Period:-

Metallic character decreases as you move across from left to right in the Periodic table. This is because effective nuclear charge increases across a period.

— 08 (Assessment Zone No:- 4-B) 80 —

Question:- Which element has lower metallic character.

i) Li or K

Sol:- Group 1

Li has lower metallic character

Reason:- As we move down the group the metallic character increases because atomic size increase & attract of nucleus & valence increases.

ii) Mg or Ca

Sol:- Mg has lower metallic character.

ii) Mg or Ca

Sol:- Mg has lower metallic character.

— (Question & Answers) —

Question No 15:- What is Reactivity? Explain its trends in the Period table.

Answer:-

Definition:- The Capability of an element to react with other elements to form new compound is called reactivity.

Trends down the group:-

Reactivity of elements generally increases as you move down the group. This is due to the increase in atomic size.

Trends along the period:-

Reactivity tends to vary across a period. Elements on the left side of a period (Group 1 & 2) are highly reactive due to their strong tendency to lose electrons & form positive ions. Elements on the right side of a period (6 & 17) are highly reactive as well but tend to gain electron to form negative ions.

Question No 16:- What is Density? Explain its trends in the Period table?

Answer:-

Definition:- It is defined as "mass per unit Volume."

Trends down the Group:-

Density of elements generally increases as you move down the group. This is due to the increase in atomic mass & the large size of atom.

Trends along the Period:-

Density can vary across the period. In general, density tends to increase from left to right until it reaches a maximum around the middle of the Period, & then it starts to decrease.

Question no 7:- What are the characteristic properties of Alkali Metals.

Answer:- Some characteristic properties of alkali metals are below:-

- i) Alkali metals are highly reactive metals. This trend increases down the group.
- ii) Alkali metals have low densities & are relatively soft. This trend increases down the group.
- iii) They have low melting points.
- iv) They are highly reactive & monovalent elements.
- v) They are ~~excellent~~ ^{good} conductors of electricity & heat.

Question no 18:- What are Lanthanides & Actinides?

Answer:- LANTHANIDES and ACTINIDES:-

"Lanthanides" also known as "rare" earth elements are a series of 14 elements located at the bottom of the Periodic table.

They include elements with atomic number 57 to 71.

"Actinides" are another series of 14 elements located just below Lanthanides. They include elements with atomic no 89 to 103.

— ::(Question and Answer):: —

Question no 19:- Define Transition elements?

Answer:-

TRANSITION ELEMENTS

Elements located in d-Block (Group 3 to 10) in the Periodic table are call Transition elements.

Some of there properties are follows as:-

High Density:-

Transition elements generally possess high densities due to their higher atomic mass. For eg:- iron (Fe) has density of 7.87 g/cm^3 tungsten (W) has a density of 19.3 g/cm^3

High melting Point:-

Transition elements have high melting point. This is because their metallic bonding is stronger.

Variable Oxidation States:-

Transition elements exhibit multiple oxidation states. This is because of d-subshell can also participate in bonding along with s-subshell. For eg:- iron (Fe) has oxidation states 2, 3, 4, 6

Coloured Compounds:-

Transition elements often exhibit vibrant colours. For eg:- copper compounds appear blue or green, chromium compounds often red or green.

Catalysts For industrial process:-

Transition ^{metals} elements & their compound are widely used as catalyst in various industrial process. For eg:-

- Iron is used in the Haber process for the synthesis of ammonia.
- Nickel is used as catalyst in the manufacture of margarine.

Platinum & palladium are used as catalyst in catalytic converters to reduce harmful emissions in automobiles & industrial units. (iv) Platinum is used as catalyst in the contact process for the manufacture of sulphuric acid.

Question no. 20: Define Halogens.

Answer: HALOGENS :- The elements in group 17 are called Halogens. The name halogens is derived from the Greek word "halos" meaning "salt" & "gen" ^{meaning} "former". Halogens include fluorine, chlorine, bromine, iodine, astatine, & tennessine (astatine & tennessine) are radio-active elements. Little is known about their properties.

All halogens are reactive non-metals & exist as diatomic molecules.

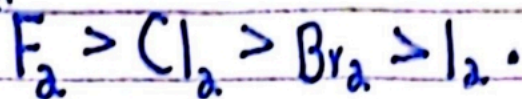
Appearance of halogens :- The colour of halogens become darker as you go down the group. At room temperature & pressure (RTP) Fluorine (F_2) exist as pale yellow gas, chlorine (Cl_2) as yellow green gas, bromine (Br_2) as red-brown liquid and iodine (I_2) as grey black solid. Iodine easily turn into a dark purple vapours on warming.

Electronic Configuration :- Halogens ^{have} 7 electrons in their valence shell. They have general electronic configuration $ns^2 np^5$. They tend to gain one electron to form univalent negative ions, F^-, Cl^-, Br^-, I^- .

Density of halogens :- As you move down the group the no. of electrons & protons increase, the size of the atoms increase & the volume increase. ^{So the density} However, which is mass per unit volume increases in general. Also fluorine & chlorine gases, bromine is a liquid & iodine is a solid.

Reactivity of halogens :- The reactivity of halogens is directly related to their ability to gain an electron & form a halide ion (fluorine ion F^- , chlorine ion Cl^- , bromide ion Br^- , iodide ion I^-). As you move down the group, the electronegativity decreases.

Because halogens have a strong tendency to gain electrons, they are a strong oxidizing power. & this power decreases down the group. Thus, the order

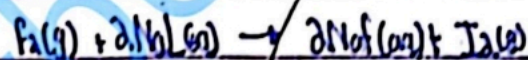
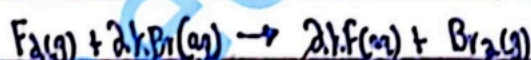


~~Exercise~~

~~Concept~~

Question no 21:- What are displacement reaction of halogens?

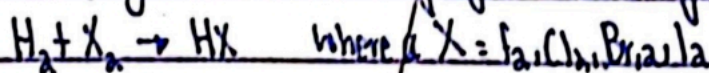
Answer:- It is possible for a free halogen to oxidize or displace the ion of halogen next to it in the group from their aqueous solutions. This means F_2 can oxidize & displace all the halide ions to free halogen. for eg:-



Similarly Cl_2 can oxidize Br^- & I^- ions. But I_2 cannot oxidize any halide ion.

Question no 22:- What are Hydrogen halides?

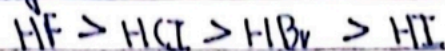
Answer:- Halogens react with hydrogen to form hydrogen halides.



The strength of the hydrogen-halogen bond is related to the electronegativity difference b/w the hydrogen & halogen atom. A larger electronegativity difference results in a stronger bond. As we move from HF to HI , the electronegativity difference b/w the hydrogen & halogens atoms. The relative thermal stability of Hydrogen halides gradually decrease from HF to HI .

As we move from HF to HI, the electronegativity difference b/w the hydrogen & halogens atoms. The relative thermal stability of hydrogen halides gradually decrease from HF to HI.

Consequently, the energy needed to break H-X decrease in the following order.



Question no 23:- What are the properties of halogens.

Answers:- The elements present in Group 17 are called halogens. They are poisonous non-metals that have low melting & boiling points that does increase down the group. (Fluorine & chlorine, the 1st & 2nd halogens are a gas, bromine, the 3rd halogen is a liquid, & the iodine, the 4th halogen is a solid.) The colours of halogens also get darker from top to bottom.

Question no 24:- What are noble gases? Write their properties.

Answers:- Noble gases, also known as inert gases, are a group of chemical elements found in Group 18 of the periodic table. They have general electron configuration ns^2, np^6 except He, which has $1s^2$. They are characterized by unique properties. They are odorless, colourless, ~~mono~~ ^{mono} atomic gases & possess very low reactivity with other elements. Noble gases include elements: Helium (He), ^{Neon} Argon (Ne), argon (Ar), krypton (Kr), xenon (Xe), radon (Rn) & oganesson (Og).

Uses:-

Due to their non-reactive nature, noble gases are used in many ways, such as in lighting (eg neon signs), refrigeration systems & welding.

Question No 25:- What are the General physical properties of metals and non-metals.

Answer:-

Thermal Conductivity:-

Metals generally have high thermal Conductivity, which means they can Conduct heat easily. On the other hand, non-metals tend to have poor Conductivity, making them less efficient at Conducting heat.

Electrical Conductivity:-

Metals are good Conductor of electricity, because they have free electrons that can move freely in the metal lattice.

Non-Metal, with few exception such as graphite, are poor Conductor of electricity because they Lack free electrons.

Adaptability:-

The metals are malleable and ductile. So they can be hammered drawn into wires or transformed into thin Sheets without breaking.

Melting & Boiling Point:-

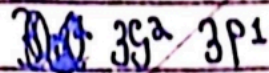
Metals generally have high melting points & boiling points due to strong metallic bond that require a lot of energy to break. On the other hand non-metal often have Lower melting points & boiling points because their atom & molecules are held by weaker bond.

— :: (Exercise Question and Answer) :: —

Q no 1. Short Q/Answer

Question no 1: Write the valence shell electronic configuration of an element present in the 3rd period & Group IIIA

Answer:-



Question no 3: Which atom has higher shielding effect Li or Na?

Answer:-

Na has higher shielding effect.

Question no 4: Explain why Na has higher ionization energy than K.

Answer: "Na" has higher energy than "K"

Reason:-

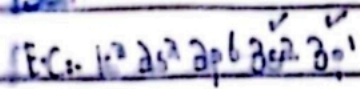
"K" comes down the "Na" in Group IA & I.E. decreases as we move down the group in P.T.

Question no 5: Alkali metals belong to S-block in the Period table, why?

Answer: Because their valence shell electronic configuration ends in S-block.

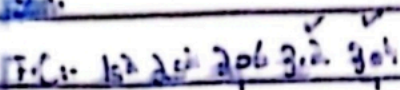
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Question no 2: In normal elements, the number of valence electrons or electrons of an element is equal to the group no. find the group no. of the following elements.



No. of valence shell = $2+1 = 3$

Group III A.



No. of valence shell = $2+4 = 6$

Group VI A.



No. of valence shell = 1

Group I A.



No. of valence shell = $2+4 = 6$

Group VI A.

Question No 9:- Write the valence shell electronic configuration for the following groups:-

a) Alkali metals.

Valence Shell electronic Configuration = ns^1

b) Alkaline earth metals.

Valence Shell electronic Configuration = ns^2

c) Halogens

Valence Shell electronic Configuration = $ns^2 np^5$

d) Noble gases.

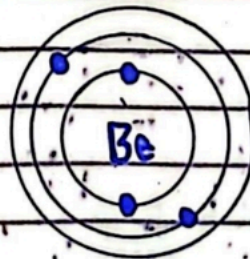
Valence Shell electronic Configuration = $ns^2 np^6$

Question No 10:- Write electron dot symbol for an atom of the following elements.

i) Be

Sol:-

atomic no is = 4



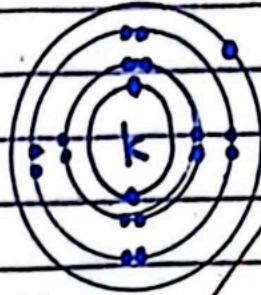
1st Shell = 2

2nd Shell = 2

ii) K

Sol:-

atomic no.: 19



1st Shell: 2, 2nd Shell: 8

3rd Shell: 8, 4th Shell: 1

iii) N

Sol:-

atomic no.: 7

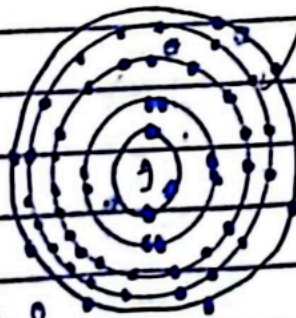


1st Shell: 2, 2nd Shell: 5

iv) I

Sol:-

atomic no.: 53



1st Shell: 2, 2nd Shell: 8

3rd Shell: 18, 4th Shell: 18, 5th Shell: 7

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Question no 11: Write the Valence Shell electronic Configuration of the atoms of the following elements.

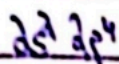
a) An element present in period 3 of Group VII

Sol:



b) An element present in period 2 of Group VII

Sol:



Question no 12: Copy & Complete the following table.

Atomic number	Mass no	No of Protons	No of neutrons	No of electrons
11	23	11	12	11
14	29	14	15	14
22	47	22	25	22
13	27	13	14	13

Question no 13: In which block, group - - - - -
- - - - - electronic Configuration.

i) $1s^2 2s^1$

= Period II Group I A.

ii) $1s^2 2s^2 2p^5$

= Period II Group VII A.

iii) $1s^2 2s^2 2p^6 3s^2$

= Period III Group II A.

iv) $1s^2$

= Period I Group VII A.

--- (Chapter Complete) ---